

Learning Conway’s Game of Life with CNN-Transformer Hybrids: Architecture Scaling, Exposure Bias, and Activation Inductive Bias

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August 6, 2026

Abstract

We study the problem of learning the one-step transition rule of Conway’s Game of Life (GoL) on a 40×40 toroidal grid using a CNN-Transformer hybrid architecture. Starting from a model trained with pure teacher forcing, we show that autoregressive rollout suffers from exposure bias: the model diverges on sparse configurations it never sees during single-step training. We show that a straight-through-estimator (STE) multi-step training scheme fails due to a systematic gradient bias, while scheduled sampling resolves the issue cleanly. We further show that once a 291K-parameter model saturates its representational capacity, scaling the training set five-fold *regresses* performance, while scaling the model five-fold to 1.5M parameters eliminates essentially all remaining errors. Motivated by recent work on activation-function inductive bias for cellular automata (Ahmed & Davis, 2026), we independently reproduce their central finding that a learnable second-degree polynomial activation lets a 25–34 parameter CNN learn GoL reliably where ReLU fails almost completely, and we report a negative result from attempting to transfer this activation into our larger multi-step model. Finally, we analyze the learned patch embeddings of our best models under geometric transformations and cellular perturbations, characterizing how content, absolute position, and attention-mediated context interact in the representation space.

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1 Introduction

Conway’s Game of Life (GoL) is a deterministic cellular automaton defined on a two-dimensional grid, where each cell is alive or dead and updates synchronously according to a fixed local rule (B3/S23): a dead cell with exactly 3 live neighbors becomes alive (*birth*); a live cell with 2 or 3 live neighbors stays alive (*survival*); all other cells die or remain dead. Despite the simplicity of the rule, GoL exhibits rich emergent dynamics (oscillators, spaceships, and chaotic regimes at high density), which makes it a useful and cheaply-simulated testbed for studying whether neural networks can learn a precisely-specified local update rule from examples, and how architectural and training choices affect that learning.

This report documents an iterative series of experiments training a CNN-Transformer hybrid to predict one GoL step ahead on a 40×40 toroidal (periodic-boundary) grid. The narrative proceeds through three largely independent threads:

1. **Training dynamics for multi-step prediction** (Section 3). A model trained purely with teacher forcing (always conditioning on the ground-truth previous frame) performs well on single-step prediction but collapses under its own autoregressive rollout, particularly on sparse configurations. We compare a straight-through-estimator (STE) approach to multi-step training against scheduled sampling, and show that only the latter is stable.
2. **Model capacity vs. data scale** (Section 4). Once a model of a given size is performing near-perfectly on average, we show that increasing the *training set size* five-fold can *regress* accuracy on hard (high-density) grids, while increasing *model capacity* five-fold removes essentially all remaining errors. This is consistent with the model having saturated its representational capacity rather than being data-limited.
3. **Activation-function inductive bias** (Section 6). Motivated by a recent paper on Kolmogorov-Arnold-style polynomial activations for GoL [1], we independently reproduce its central claim on our own 40×40 toroidal setup using an from-scratch reimplementation of their minimal architecture, and separately report a negative result from attempting to transfer the same activation into our larger multi-step CNN-Transformer.

Section 2 describes the shared CNN-Transformer architecture used throughout. Section 5 presents an interpretability study of the learned patch embeddings under geometric transformations (rotation, reflection, translation) and local perturbations (single-cell flips and moves), aimed at understanding how content, positional encoding, and attention-mediated context jointly determine the representation of a grid patch.

2 Architecture

All models in Sections 3–5 share the same *CNN-Transformer* architecture (referred to as **CNNTransformerV4** in the codebase), operating on a 40×40 binary grid with toroidal (periodic) boundary conditions.

The forward pass has three stages:

Stage 1 — Local CNN feature extractor. Two 3×3 convolutions with **circular** padding mode (so that convolution respects the toroidal topology exactly, rather than treating the grid border as a hard edge), each followed by GroupNorm and a GELU nonlinearity:

$$\text{Conv2d}(1 \rightarrow 32) \rightarrow \text{GroupNorm} \rightarrow \text{GELU} \rightarrow \text{Conv2d}(32 \rightarrow d_{\text{model}}) \rightarrow \text{GroupNorm} \rightarrow \text{GELU}.$$

The receptive field of this stage is 5×5 , matching (and slightly exceeding) the Moore neighborhood relevant to the GoL update rule.

Stage 2 — Patch tokenization. The 40×40 feature map is partitioned into non-overlapping 4×4 patches (a 10×10 arrangement, 100 patches total), following the patch-tokenization scheme of Vision Transformers [3]. Each patch’s $d_{\text{model}} \times 4 \times 4$ feature block is flattened and linearly projected to a d_{model} -dimensional token, and a *learnable* absolute positional embedding $\text{pos_embed} \in \mathbb{R}^{100 \times d_{\text{model}}}$ is added:

$$\text{token}_i = \text{Linear}(\text{feat}_i) + \text{pos_embed}_i, \quad i = 1, \dots, 100.$$

We refer to this as the **pre-transformer embedding** in Section 5.

Stage 3 — Transformer encoder. A pre-norm Transformer encoder [5] (GELU activation, $\text{dim_feedforward} = 4d_{\text{model}}$) mixes information across all 100 patch tokens via self-attention. We refer to its output as the **post-transformer embedding**. A final linear head maps each output token back to $4 \times 4 = 16$ logits, reconstructing next-state predictions for every cell in the patch.

Model sizes used. Two sizes appear throughout this report:

Name	d_{model}	heads	layers	dim_feedforward	params
V8 (baseline size)	64	4	4	256	291,888
V10 (scaled size)	128	8	6	512	1,504,240

Data augmentation. Training uses the dihedral group D_4 (90° rotations and horizontal/vertical reflections) applied jointly to input and target frames. Translation augmentation is deliberately *not* used, since it directly conflicts with the learnable absolute positional embedding in Stage 2 (a translated patch would need a different position embedding than the one it is assigned).

3 Training Dynamics for Multi-Step Prediction

3.1 Baseline: single-step teacher forcing (V5)

The V5 model is trained with plain single-step supervision: at every optimization step, the model observes the ground-truth frame at time t and is trained (binary cross-entropy with label smoothing $\varepsilon = 0.1$) to predict the ground-truth frame at $t + 1$. This is standard *teacher forcing* and never exposes the model to its own prediction errors during training.

V5 reaches strong single-step metrics (best validation loss 0.20125, precision 99.4%, recall 99.7%), but per-event accuracy reveals a weakness: *born* accuracy (correctly predicting a birth, i.e. dead→alive) is only 93.8%, well below *survival* accuracy (99.8%) and *death* accuracy (99.9%). When the trained model is unrolled autoregressively (feeding its own prediction back in as the next input,

rather than the ground truth), sparse configurations tend to collapse to the all-dead state after a handful of steps: because the model was never trained on grids containing its own error patterns, occasional missed births are never corrected and instead compound, until the local neighborhood has too few live cells to sustain any further births. This is the classic *exposure bias* problem for autoregressive sequence models.

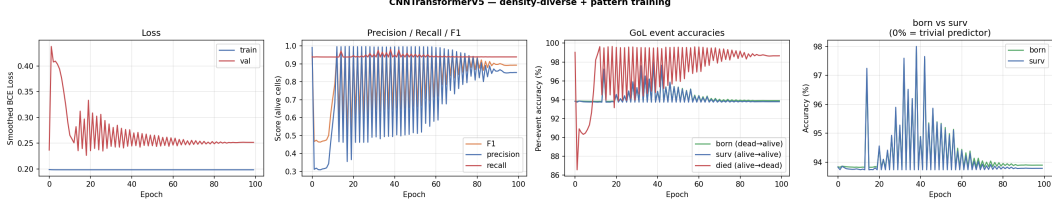


Figure 1: V5 training curves (single-step teacher forcing). Aggregate metrics look strong, but this model still collapses under autoregressive rollout on sparse grids.

3.2 Failed fix: straight-through estimator multi-step training (V6)

To directly train for rollout stability, V6 unrolls the model for $K = 3$ steps during training. At each step, the model’s *hard* (binarized) prediction is fed forward as the next input, while gradients are passed back through the *soft* (sigmoid) prediction via a straight-through estimator (STE) — i.e., forward = $\mathbb{I}[\sigma(x) > 0.5]$, backward = $\nabla\sigma(x)$.

V6 performs well only transiently: at epoch 5 (warm-started from V5) it reaches validation loss 0.23421 with precision 92.7% and born-accuracy 99.5%, but precision then collapses to 20–50% over the following epochs and oscillates chaotically through epoch 50 (Figure 2). Gradient clipping (already active at $\text{max_norm} = 1.0$) does not resolve this. We hypothesize the failure mode is a *systematic directional gradient bias*: when the model misses a birth at step t , step $t + 1$ ’s input is missing a live cell relative to ground truth, and the STE gradient at step $t + 1$ consistently signals “predict more alive” back through the network, regardless of whether more alive cells are actually correct at t . This creates a positive-feedback loop toward over-predicting liveness that is not a magnitude problem (which clipping could fix) but a *direction* problem.

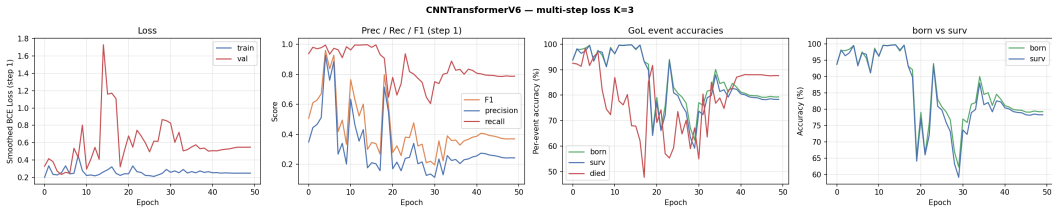


Figure 2: V6 training curves (STE multi-step, $K = 3$). Precision collapses after an initial promising region around epoch 5 and never recovers.

A second attempt with STE at a lower learning rate ($1e-5$) and $K = 2$ still diverged, with the same directional signature (precision falling monotonically from epoch 1 to epoch 5), confirming the bias is intrinsic to the STE gradient path rather than an optimization-stability artifact.

3.3 Working fix: scheduled sampling (V7)

V7 replaces the STE hard/soft split with *scheduled sampling* [2]: at each of the $K = 3$ unrolled steps, the next input is the model’s own (hard, binarized) prediction with probability p , or the ground-truth frame with probability $1 - p$, where p increases linearly from 0 to 1 over training. Critically, gradients flow only through the *current* step’s logits — there is no gradient path connecting step $t + 1$ ’s loss back to step t ’s prediction, which removes the STE’s compounding directional bias entirely.

V7 trains stably for all 50 epochs (Figure 3) and reaches the best aggregate metrics of any single-step-warm-started model in this study: validation loss 0.20114, F1 0.9985, precision 99.7%, recall 100%, and *all three* per-event accuracies (born, survival, death) at 99.7–100%. Qualitatively, the artifact of rollouts settling into sparse grid-aligned dead patterns — visible in V5/V6 rollouts on sparse inputs — disappears entirely.

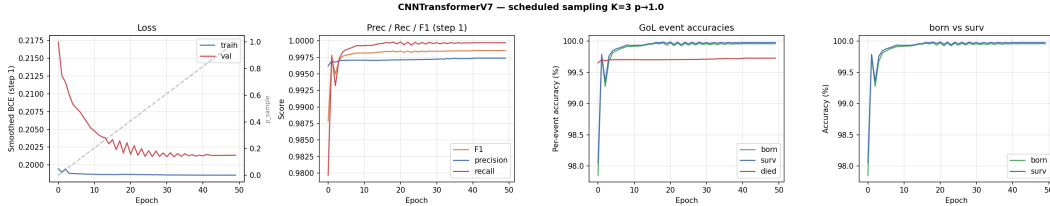


Figure 3: V7 training curves (scheduled sampling, $K = 3$, $p : 0 \rightarrow 1$). Training is stable across all 50 epochs, with monotonically improving per-event accuracies.

3.4 Targeted fine-tuning at high density (V8)

Although V7’s aggregate metrics are strong, a targeted failure search (random grids across densities 0.02–0.80, and 8 named small patterns, seed-matched for reproducibility) finds that essentially all remaining errors concentrate at high density ($d = 0.80$), with roughly 100 false positives per grid on average.

V8 continues training V7’s checkpoint for 30 further epochs with a dataset that upsamples $d = 0.65$ and $d = 0.80$ trajectories by $3\times$ relative to V7’s dataset, while retaining the full range of lower densities (so the model does not lose accuracy on sparse/medium configurations it already handles well). This reduces the mean false-positive count at $d = 0.80$ from approximately 100/grid to approximately 4.6/grid — a $21\times$ improvement — while aggregate metrics improve marginally (validation loss 0.20064, precision 100%, recall 99.9%). Figure 4 shows a direct comparison of V7 and V8 on one of the exact failure grids identified in the V7 failure search.

3.5 Data scale-up regresses performance (V9)

Given V8’s improvement from more high-density data, we test whether scaling *all* data five-fold (approximately 546,200 total training examples, warm-started from V8) improves matters further. Aggregate validation metrics remain similar to V8 (validation loss 0.20118, precision 99.7%, recall 99.6%), but per-grid error counts on the same failure-search protocol *worsen substantially*: false positives at $d = 0.80$ increase from V8’s 2,296 (across 500 grids) to 22,190, and false negatives at $d = 0.50$ increase from 67 to 6,948.

This is the central negative result motivating Section 4: *aggregate validation metrics can look stable even as tail-case behavior regresses sharply*, and simply scaling the training set does not help once a model has saturated its representational capacity — indeed it can actively hurt,

V7 vs V8 — failure case 1 d0.80 v7err107
V7: 101 FP + 6 FN = 107 errors V8: 4 FP + 5 FN = 9 errors

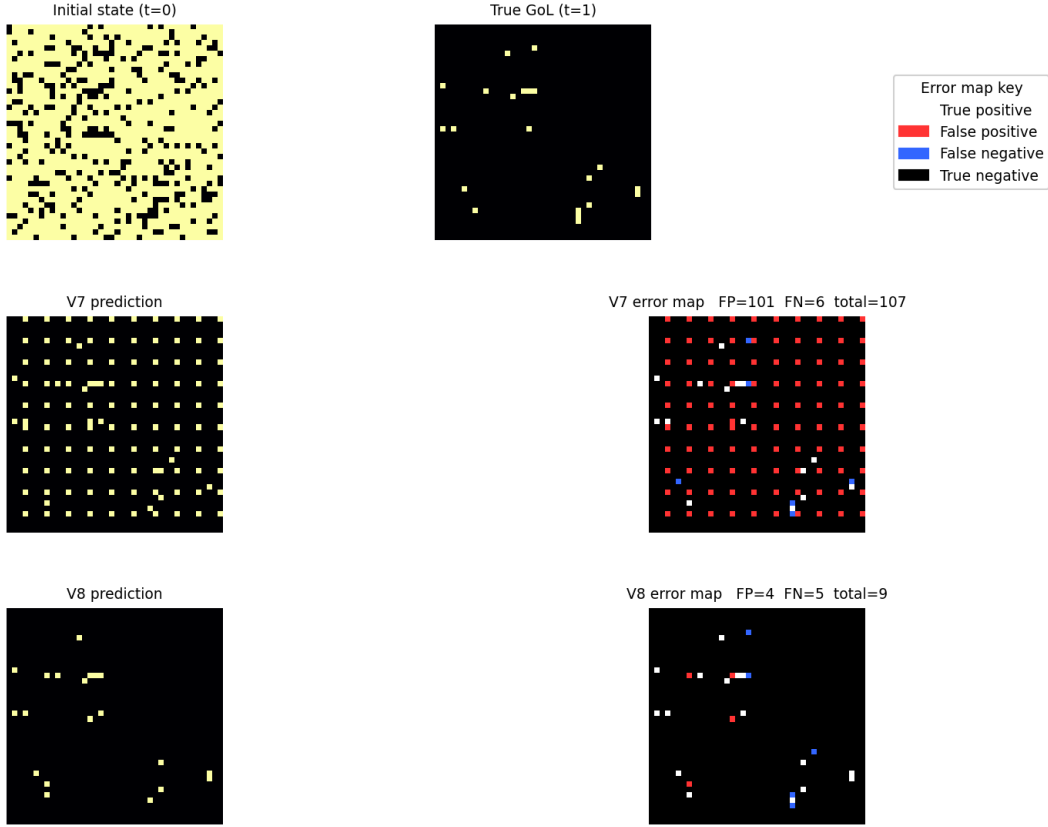


Figure 4: V7 vs. V8 on an identical $d = 0.80$ failure grid found during the V7 failure search. V7 makes 107 errors (mostly false positives, visible as a diffuse grid-aligned speckle pattern); V8, fine-tuned with $3\times$ more high-density data, makes substantially fewer errors on the same input.

plausibly because the effective gradient signal from over-represented high-density trajectories (each contributing many alive-cell examples) skews optimization away from the sparse/medium regime that was previously well-fit.

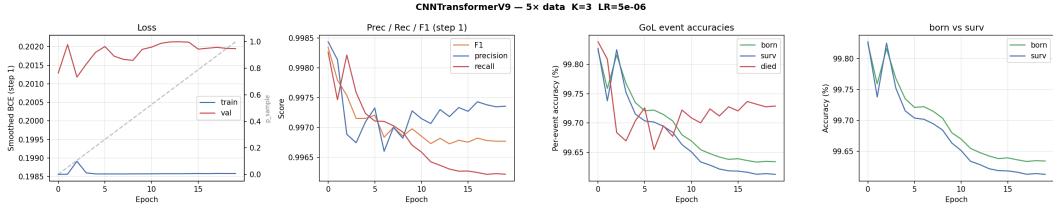


Figure 5: V9 training curves ($5\times$ data scale-up, warm-started from V8). Aggregate metrics look unremarkable, masking a substantial regression in per-grid tail-case error counts (see Table 1).

3.6 Scaling model capacity instead (V10)

Rather than scaling data further, V10 scales the *model*: doubling d_{model} ($64 \rightarrow 128$), doubling attention heads ($4 \rightarrow 8$), and adding two Transformer layers ($4 \rightarrow 6$), for a total of 1,504,240 parameters ($5.15\times$ V8’s size). Because the embedding dimension changes, V10 is trained *from scratch* (V8/V9’s weights cannot be warm-started into a different-width model) using the same scheduled sampling recipe and V8’s dataset configuration (i.e., *not* the $5\times$ scaled-up V9 dataset).

V10 converges essentially immediately: by epoch 10 of 100, all metrics (precision, recall, born, survival, death) reach 100.0% and validation loss reaches 0.19852 — notably *below* the theoretical label-smoothing floor achieved by any previous version, and the model holds this exactly through epoch 100 (Figure 6).

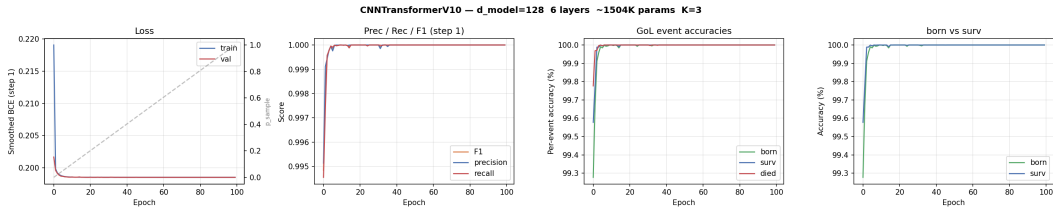


Figure 6: V10 training curves ($d_{\text{model}} = 128$, 6 layers, 1.5M params, trained from scratch). All metrics saturate at 100% by epoch 10 and remain stable through epoch 100.

Repeating the exact per-grid failure-search protocol used for V8/V9 (500 grids per density, densities 0.20–0.80, fixed seed), V10 makes **zero errors across all 2,500 evaluated grids** — a qualitative change from V8’s thousands of residual errors at high density (Table 1, Figure 7).

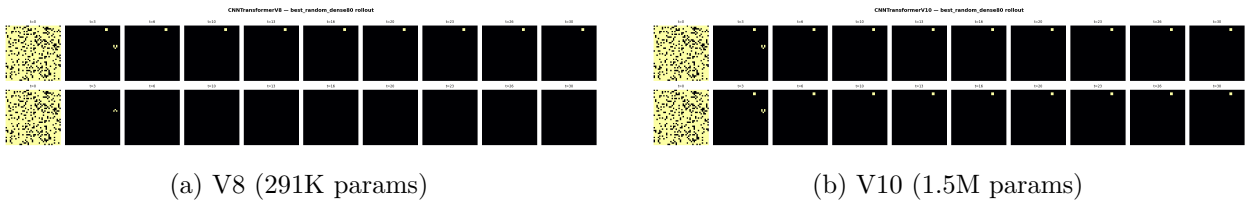


Figure 7: 30-step autoregressive rollout starting from a random $d = 0.80$ grid, true GoL trajectory (top row of each panel) vs. model prediction (bottom row).

Taken together, V8 through V10 demonstrate that *model capacity*, not data volume, was the binding constraint at the 291K-parameter scale: more data of the same kind that a saturated model already sees plenty of does not help (and can actively hurt), while more parameters directly resolve the remaining tail-case errors.

4 Summary of All Versions

Table 1 summarizes the full progression of models discussed in Section 3. “Per-grid errors” figures (where available) come from a fixed failure-search protocol: 500 random grids at each of five densities (0.20, 0.35, 0.50, 0.65, 0.80), same random seed across models, counting total false positive + false negative cells summed over all 500 grids at each density.

Ver.	Key change	Best val loss	Prec.	Rec.	Born	Status
V5	Single-step teacher forcing (baseline)	0.20125	99.4%	99.7%	93.8%	Collapses under rollout
V6	+STE multi-step ($K = 3$)	0.23421 ^a	92.7% ^a	—	99.5% ^a	Failed — diverges
V7	STE $\rightarrow$ scheduled sampling	0.20114	99.7%	100%	100%	Stable, strong
V8	+3 $\times$ high-density upsample	0.20064	100%	99.9%	99.9%	Best per-grid accuracy (291K)
V9	+5 $\times$ all data	0.20118	99.7%	99.6%	99.6%	Regressed per-grid errors
V10	5.15 $\times$ model size, from scratch	0.19852	100%	100%	100%	Zero errors , 2,500/2,500 grids

Table 1: Version summary. ^aV6’s best epoch (5) before divergence; the model never recovers after this point.

Density	V8 total errors	V9 total errors	V10 total errors
0.20	0	—	0
0.35	192	—	0
0.50	94	6,948 ^b	0
0.65	1,891	—	0
0.80	3,358	22,190	0

Table 2: Per-grid error counts (false positives + false negatives, summed over 500 grids per density) for the three model sizes discussed in Section 3. ^bV9 figure is false negatives specifically, quoted as reported during the original failure comparison; full V9 error breakdown at other densities was not re-collected after the regression was identified and model capacity scaling (V10) was pursued instead.

The central empirical claim of this section is visible directly in Table 2: V9’s aggregate validation metrics (Table 1) are statistically indistinguishable from V8’s, yet its per-grid tail-case error count is an order of magnitude *worse*. This discrepancy would not have been visible from validation loss or aggregate precision/recall alone — it required a targeted, density-stratified failure search to detect. We consider this the main practical lesson of the V8–V10 progression: *aggregate metrics on a held-out validation set with the same distribution as training data can silently hide severe regressions in specific, identifiable subpopulations of the input space*, and per-subgroup error auditing is necessary to catch this class of regression.

5 Embedding Analysis

To understand what the learned patch representations encode, we extract two embedding tensors per grid from the V8 and V10 models (Section 2): the **pre-transformer** embedding (patch content projection + learnable positional embedding, before attention) and the **post-transformer** embedding (after the Transformer encoder mixes information across all 100 patches). Both are $100 \times d_{\text{model}}$ tensors — one d_{model} -dimensional vector per 4×4 patch, arranged on a 10×10 grid.

5.1 Static structure: patch norms and pairwise similarity

For a fixed configuration, we visualize (i) the spatial map of post-transformer embedding norms, $\|\text{emb}_i\|_2$ for each of the 100 patches, and (ii) the full 100×100 pairwise cosine similarity matrix between all patches’ embeddings.

Figure 8 shows that for sparse hand-placed patterns, roughly 95–98 of the 100 patches are entirely empty and produce near-identical embeddings (a large, uniformly high-similarity block in the pairwise matrix), while the handful of occupied patches form a small, distinct cluster. Figure 9 shows this structure eroding as density increases: at $d = 0.80$, essentially every patch has unique content, and the similarity matrix loses its block structure entirely. This is a direct, visualizable signature of the sparse/dense distinction that motivated the density-targeted fine-tuning in Section 3 (V8).

5.2 Sensitivity to transformations

We next measure how a patch’s embedding changes under four families of transformation applied to a fixed base configuration (glider, blinker, or beacon): (i) $90^\circ/180^\circ/270^\circ$ rotation, (ii) horizontal/vertical reflection, (iii) toroidal translation by 1 or 4 cells, and (iv) local perturbation — either flipping a single cell’s state, or moving a single live cell to an adjacent position. For each transformation we compute $\Delta = \text{emb}(\text{transformed}) - \text{emb}(\text{original})$ per patch, both pre- and post-transformer.

Table 3 (mean cosine similarity between original and transformed post-transformer embeddings, averaged over all 100 patches) quantifies the pattern visible in Figure 10: sparse patterns (glider) are nearly invariant to all transformations (cosine similarity $\gtrsim 0.99$, since almost every patch is empty on both sides of the transformation and empty patches encode almost identically regardless of the grid’s global content), while random grids show much larger sensitivity to rotation, reflection, and coarse translation (≈ 0.75 – 0.92) because *every* patch’s absolute position changes and the learnable positional embedding is position-specific, not translation- or rotation-equivariant.

5.3 Cross-transformation structure and geometry

To ask whether different transformations perturb the embedding space in *similar directions*, we flatten each transformation’s full $\Delta \in \mathbb{R}^{100 \times d_{\text{model}}}$ vector and compute pairwise cosine similarity across all transformations (Figure 11). We additionally project the 100 patch embeddings (original and every transformed version) into the top-2 principal components of the original configuration’s embedding set (Figure 12), to visualize how occupied patches move through representation space under each transformation while the empty-patch cluster remains essentially fixed.

Qualitatively, geometric transformations (rotation, reflection) that share the dihedral symmetry group used in data augmentation (Section 2) tend to produce more mutually correlated Δ vectors than translation does, consistent with the model having been explicitly trained to be robust to

V8 Embeddings — Oscillators

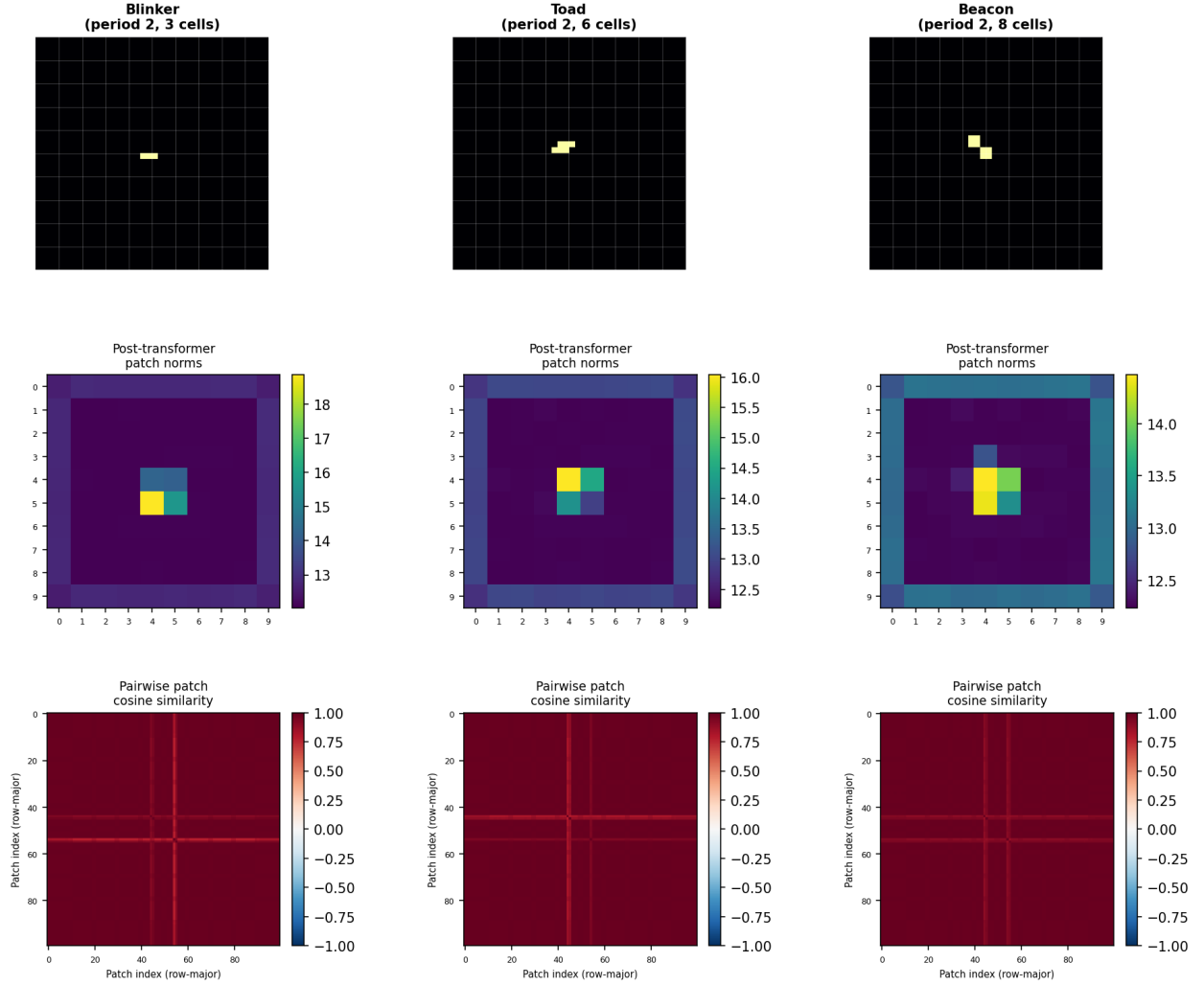


Figure 8: V8 embeddings for three period-2 oscillators (blinker, toad, beacon). Top row: grid configuration. Middle row: post-transformer patch norm heatmap. Bottom row: 100×100 pairwise cosine similarity matrix. Sparse configurations produce a large, near-uniform block of mutually similar “empty-patch” embeddings, with only the 1–2 occupied patches standing apart.

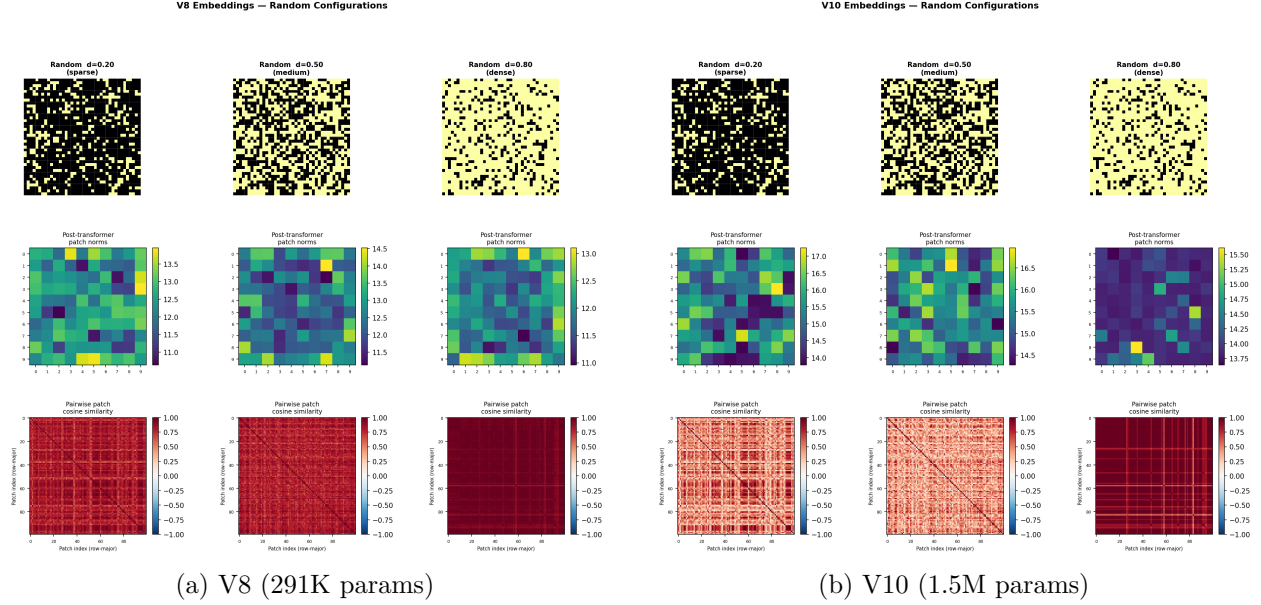


Figure 9: Embeddings for random configurations at densities 0.20, 0.50, 0.80 (left to right within each panel). As density increases, every patch carries genuine content, and the large “empty-patch” block seen in Figure 8 disappears, replaced by a more heterogeneous similarity structure.

Transformation	Glider	Random ($d=0.35$)	Random ($d=0.65$)
Rotate 90°	0.992	0.751	0.916
Rotate 180°	0.991	0.744	0.909
Flip horizontal	0.991	0.764	0.905
Flip vertical	0.992	0.758	0.911
Translate +4 cols	0.993	0.763	0.905
Translate +1 col	0.996	0.780	0.924
Cell flip (alive→dead)	0.9996	—	—
Cell flip (dead→alive)	0.9988	—	—
Cell move (1 step)	0.9987	—	—

Table 3: Mean post-transformer cosine similarity between original and transformed embeddings, V8. Sparse configurations are nearly invariant to all transformations; dense random grids are substantially more sensitive to geometric transforms, because the learnable absolute positional embedding breaks exact translation/rotation equivariance.

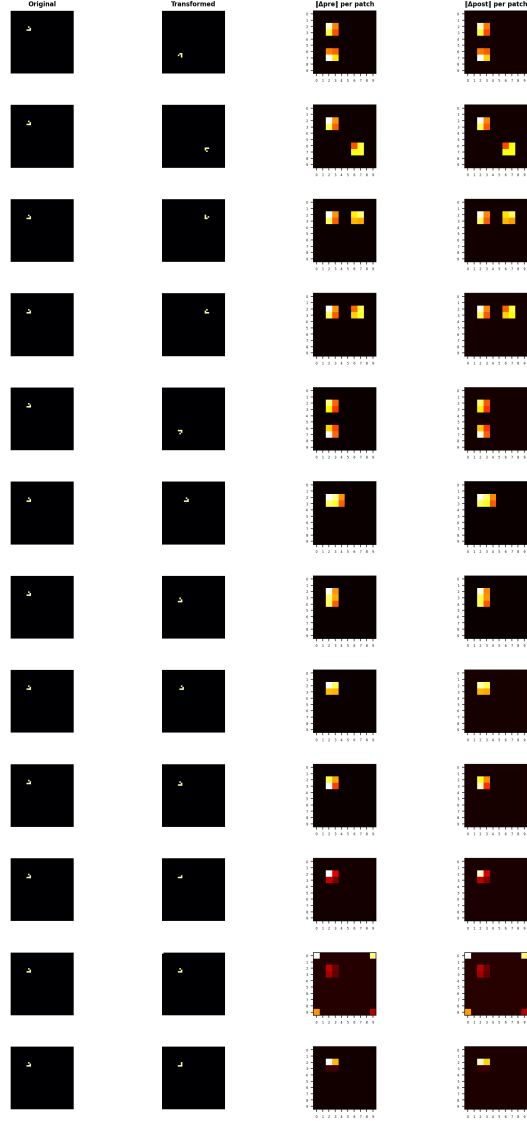


Figure 10: V8: per-patch embedding change $\|\Delta\|_2$ (pre- and post-transformer) for a glider under each transformation. Geometric transforms (rotation, reflection, coarse translation) perturb many patches at once because the pattern relocates to different absolute positions; single-cell perturbations affect only the one patch containing the perturbed cell.

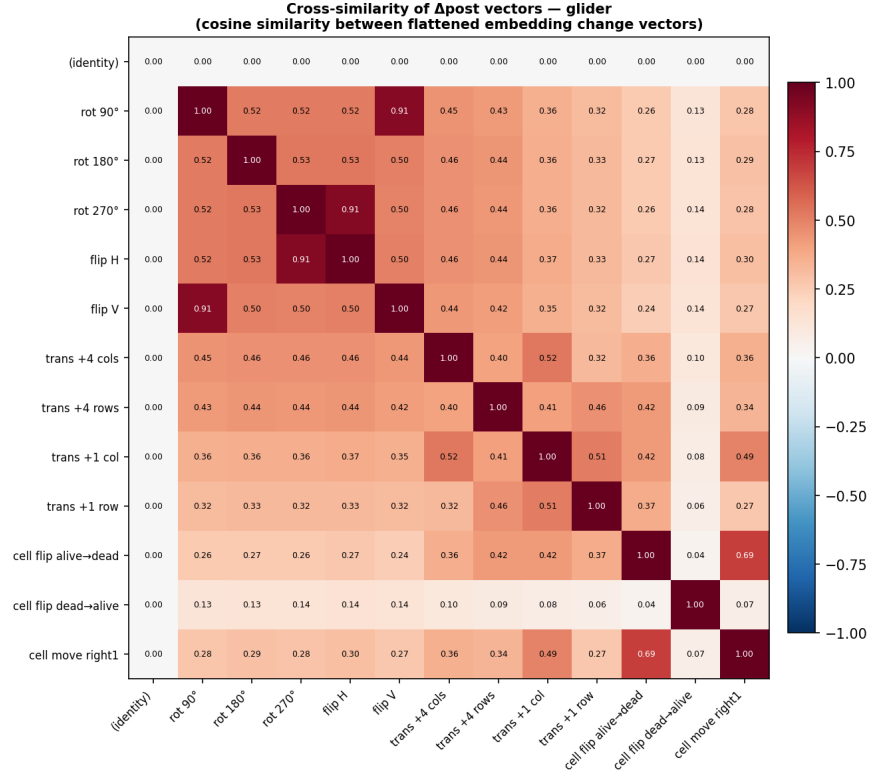


Figure 11: V8: cosine similarity between the flattened Δ_{post} vectors of every pair of transformations (glider base pattern). Values near 1 indicate two transformations perturb the embedding space in nearly the same direction; values near 0 indicate unrelated perturbation directions.

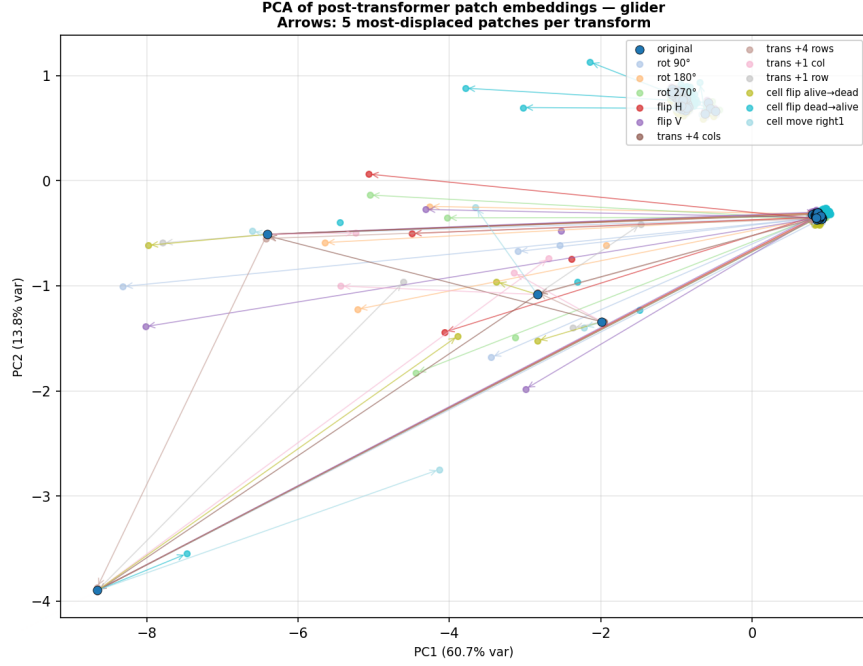


Figure 12: V8: PCA projection (fit on the original glider’s 100 patch embeddings) of the embeddings under every transformation. The large central cluster is the ~ 95 empty patches, essentially stationary across all transformations; arrows trace the 5 most-displaced patches per transformation, showing where the occupied-patch information moves to under each transform.

(though not embedding-invariant under) D_4 transformations, while translation invariance was never encouraged during training.

6 Activation-Function Inductive Bias

6.1 Motivation: Polynomial Kolmogorov-Arnold Networks

Ahmed & Davis [1] study why small neural networks struggle to *learn* the GoL transition rule despite minimal networks being known to *represent* it exactly with only a handful of hand-set parameters [4]. They argue that the GoL update is a non-monotonic function of live-neighbor count N — flat at zero for $N \in \{0, 1\}$, rising sharply for $N \in \{2, 3\}$, and flat at zero again for $N \geq 4$ — which a monotonic activation such as ReLU can only approximate piecewise, requiring many units and a fortunate initialization (a “winning ticket”) to succeed reliably. They show that replacing ReLU with a *learnable second-degree polynomial* activation,

$$\phi(x) = w_0 + w_1x + w_2x^2,$$

lets a minimal CNN — one 3×3 circularly-padded convolution, one 1×1 convolution, and a final 1×1 convolution with sigmoid output — learn GoL reliably on a 32×32 toroidal grid, whereas the same architecture with ReLU activation fails almost completely.

6.2 Independent reproduction

Before attempting to transfer this idea into our larger CNN-Transformer, we independently verify the paper’s central claim on our own 40×40 toroidal setup, using a from-scratch reimplementaion (available in `poly_activation_verify/`, kept deliberately separate from the main codebase) with no dependency on our own GoL simulator or model code. We build the minimal $L(1, m=1)$ architecture described above and train two variants — identical in every respect except the activation function used at the two hidden activation sites — with plain single-step teacher forcing (binary cross-entropy, Adam, learning rate 10^{-3} , batch size 8) on a small dataset of 100 training / 20 validation random grids (density sampled uniformly in $[0.1, 0.9]$), across 10 random seeds per activation.

Activation	Parameters	Success rate	Mean epochs to converge	Mean final accuracy
ReLU	25	1/10	159.0 (of successful run)	89.39%
Polynomial	34	10/10	185.6	100.00%

Table 4: Reproduction of Ahmed & Davis [1] on our own 40×40 toroidal grid and independent implementation. Parameter counts (25 and 34) match the paper’s reported minimal-network sizes exactly. “Success” is defined as reaching $\geq 99.99\%$ validation cell accuracy for two consecutive epochs, within 500 epochs.

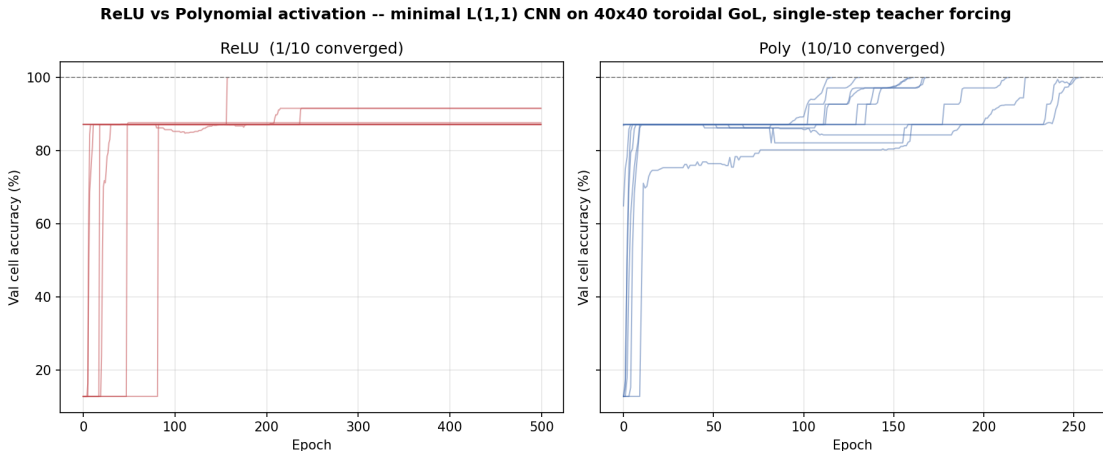


Figure 13: Validation accuracy over training, all 10 seeds. ReLU (left) plateaus around 87–92% for 9 of 10 seeds and never escapes; polynomial activation (right) reaches 100% accuracy on every seed.

The reproduction (Table 4, Figure 13) confirms the paper’s claim under an independent implementation, a different grid size, and a different (and much smaller) training set than the original paper used: ReLU converges to perfect accuracy on only 1 of 10 seeds, plateauing at 87–92% cell accuracy on the rest, while the polynomial activation converges to perfect accuracy on all 10 seeds, typically within 120–260 epochs (each run completing in 1–2 seconds on CPU, given the network’s tiny size).

6.3 Negative result: transferring the polynomial activation to the multi-step model

Given the clean reproduction in Section 6.2, we attempted to transfer the polynomial activation into the CNN feature-extraction stage of the full CNN-Transformer architecture (Section 2), at V8’s parameter budget ($d_{\text{model}} = 64$, 4 layers): replacing the two GELU activations in Stage 1 with the same learnable $\phi(x) = w_0 + w_1x + w_2x^2$ used in Section 6.2, while leaving the Transformer encoder’s GELU unchanged. This adds only 288 parameters over the baseline V8 architecture (292,176 vs. 291,888), so any accuracy change can be attributed to the activation rather than to capacity. The model was trained from scratch with the same scheduled sampling recipe used for V10 (Section 3): $K = 3$ unrolled steps, $p : 0 \rightarrow 1$ over training, learning rate 10^{-4} .

Training was *not* stable. By epoch 5, validation metrics were promising (F1 = 0.89, born-accuracy 86.8%, validation loss 0.227) — already close to V8’s converged performance after only 5 epochs from scratch. However, by epoch 10 the model had regressed sharply (validation loss 0.491, F1 0.39), and by epoch 15 it had diverged further (validation loss 1.045, F1 0.17), while per-epoch wall-clock time grew roughly five-fold over the same interval. Training loss remained flat throughout (≈ 0.21) even as validation loss exploded, indicating the divergence is specific to the clean single-step evaluation distribution rather than a general optimization failure.

We hypothesize this reflects an interaction between the polynomial activation and the multi-step scheduled-sampling training loop that is absent from the single-step setup verified in Section 6.2: the w_2x^2 term is unbounded and, unlike a saturating activation such as GELU, has no mechanism to suppress large intermediate activations. Once scheduled sampling begins feeding the model’s own (initially imperfect) predictions back as input to later unrolled steps, any activation values that drift into a regime where the quadratic term dominates are amplified rather than clipped, compounding across the $K = 3$ steps in a manner structurally similar to the STE failure mode identified for V6 (Section 3). We did not observe this instability in Section 6.2 because that experiment uses pure single-step teacher forcing with no autoregressive feedback loop.

This negative result was not pursued further within the scope of this report; we record it here as we consider the failure mode informative in its own right, and as a caution against assuming an inductive-bias fix verified in a minimal single-step setting will transfer unmodified to a larger multi-step training regime. Candidate fixes — left to future work — include bounding the quadratic coefficient (e.g. a saturating variant $\phi(x) = w_0 + w_1x + w_2x^2/(1 + x^2)$), using a separate, smaller learning rate for the polynomial coefficients, or first verifying single-step (teacher-forcing-only) training at V8/V10 scale before introducing scheduled sampling.

7 Discussion and Conclusion

Three broad lessons emerge from this study.

Exposure bias is a first-order concern for autoregressive cellular-automaton models, and its fix must be chosen carefully. Teacher forcing alone (V5) produces excellent single-step metrics that do not predict rollout stability. The most direct-seeming fix for multi-step training — a straight-through estimator (V6) — fails due to a systematic directional gradient bias rather than an optimization-stability issue, and is not rescued by gradient clipping or a lower learning rate. Scheduled sampling (V7), which avoids passing gradients across unrolled steps entirely, resolves the problem cleanly. This distinction — between instabilities caused by gradient *magnitude* and those caused by gradient *direction* — recurred later in the polynomial-activation negative result (Section 6.3), where we again observed a divergence with a similar signature (stable training loss,

exploding validation loss) under a different mechanism (an unbounded activation interacting with autoregressive feedback rather than an STE gradient path).

Aggregate validation metrics can hide severe, structured regressions. The V8-to-V9 comparison (Section 3) is the clearest demonstration: V9’s validation loss and precision/recall are statistically indistinguishable from V8’s, yet its per-grid error count at high density is an order of magnitude worse. This was only detectable via a targeted, density-stratified failure search — a general reminder that i.i.d. validation splits can fail to surface regressions concentrated in identifiable input subpopulations, and that practitioners should audit per-subgroup behavior rather than relying on a single scalar validation metric, especially before deciding that a change (like scaling up training data) has helped.

Once a model is capacity-limited, only capacity fixes the problem. The clearest evidence for this is negative: V9 attempted to resolve V8’s residual high-density errors purely with more data (of a kind the model already saw extensively) and made things worse. V10 held the data distribution fixed and scaled the model $5.15\times$, and eliminated all 2,500 evaluated per-grid errors. This is consistent with, though not identical to, the picture in Ahmed & Davis [1]: they show that a poorly-matched inductive bias (ReLU) can make a network fail to learn a rule it is representationally capable of encoding with very few parameters, while we show that once a specific *well-matched* architecture (GELU CNN-Transformer with scheduled sampling) has saturated its capacity at a given width, adding width is what resolves the remaining gap — not more data of the same kind. Our attempt to combine both ideas (Section 6.3) was unsuccessful in its first iteration, underscoring that these two levers do not necessarily compose for free, particularly once autoregressive training is involved.

The embedding analysis (Section 5) provides a mechanistic complement to the accuracy results: it shows directly, via patch-level cosine similarity and PCA projections, that the model’s learned absolute positional embedding is responsible for most of its sensitivity to geometric transformations on dense grids, while sparse configurations are nearly embedding-invariant simply because most patches are empty on both sides of any transformation. This is consistent with the architectural choice (Section 2) to augment with the dihedral group D_4 but not with translation, since translation augmentation would directly conflict with a learnable absolute positional embedding.

Limitations and future work

This report reflects a single training run per configuration for the CNN-Transformer experiments (V5–V11), aside from the deliberately multi-seed reproduction in Section 6.2; conclusions about V6/V9/V11 instability would be strengthened by repeating those runs across seeds, as we did for the minimal-CNN verification. The V11 negative result (Section 6.3) was not root-caused beyond the hypothesis offered; a natural next step is to test the polynomial activation in a single-step-only (teacher-forcing) training regime at V8/V10 model scale, isolating whether the instability is intrinsic to the activation at this model size or specific to its interaction with scheduled sampling. The per-grid failure-search protocol used throughout (Section 3) covers five densities and a small set of named patterns; a more exhaustive audit (finer density grid, targeted adversarial search) might reveal further tail-case regressions not captured here, including for V10 despite its zero-error result on the 2,500 grids tested.

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